



IVVIA

The Fourth Way To Buy and Invest in Real Estate

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I. Intro

IVVIA is a novel approach to real estate acquisition through tokenisation and decentralised finance (DeFi). The term "to ivviate" has been coined to describe the process of gradually buying out a tokenised property. The ivviator can progressively acquire tokens, eventually becoming the sole owner—similar to a mortgage but without the burdens. Meanwhile, ivviatees (investors) profit by selling their tokens at market value and collecting rental income. This process begins with creating a trust fund, raising investment to purchase the property, and tokenising it via the IVVIA smart contract.

II. Concept

A. Quarta Via

Traditionally, there are three main ways to acquire property: purchasing with cash, buying with borrowed money (i.e., taking out a mortgage), or leasing. However, each of these options has its drawbacks. Paying cash upfront is not feasible for many due to the significant amount of money required. Leasing is a temporary acquisition of property without ownership, but it provides accommodation when other options are unavailable. The mortgage is accessible to those with a stable income, it often feels like a forced step. Mortgages can be stressful not only because of the severe consequences of default, which loom like the Sword of Damocles for decades, but also due to more routine issues. For instance, you might need to relocate to another city for a new job or because of disruptive neighbours. Selling a mortgaged property and acquiring a new one doubles all related expenses, making the process burdensome. As a result, many people endure these challenges while waiting for the distant moment of mortgage relief.

What if there was a fourth way - quarta via² (IV via)? An alternative to obtaining full ownership of a property, similar to paying a mortgage but without the associated burdens. An easy way to embark, gradually pay out and get the ownership or exit by selling the share without losing on paperwork costs.

B. The scheme

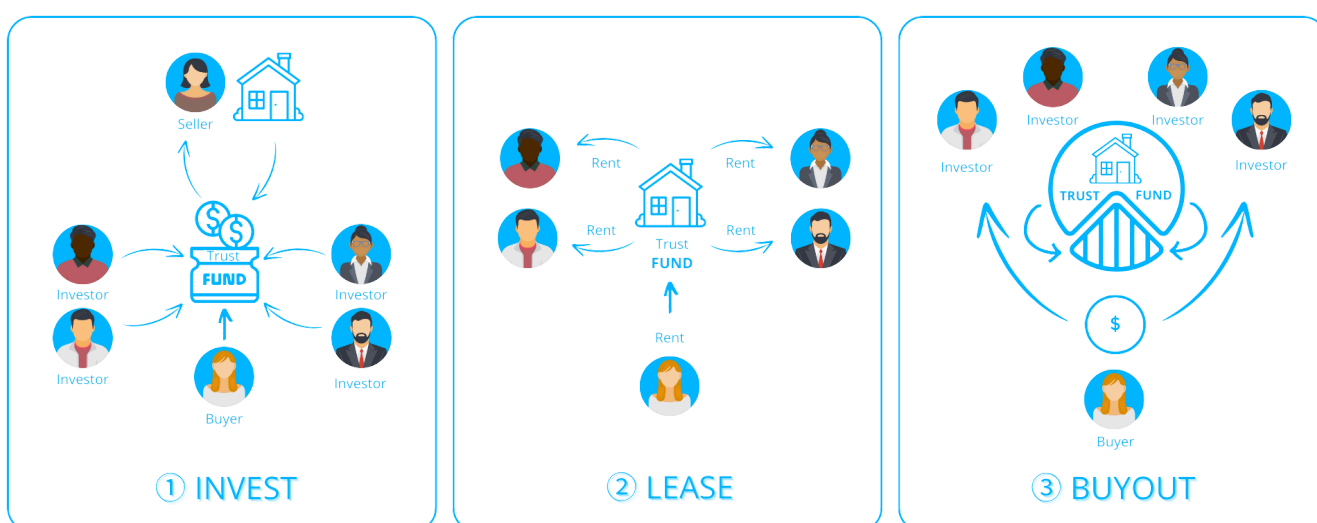
There are two key participants in this scheme: the ivviator and the ivviatee (or ivviatees). Both parties invest in acquiring the property, with the ivviator covering the down payment and the ivviatees contributing the remaining amount to match the targeted property price. To facilitate this, they form a legal entity and become its members. All the governing rules of the scheme are hardcoded in the IVVIA's smart contract, which is published on a blockchain.

To legitimise the fund, it should be formed as a trust or incorporated as a legal entity, i.e., a Special Purpose Vehicle (SPV). Their fund might also require a financial licence.

The ivviator's down payment serves as an initial contribution, signalling to ivviatees that the SPV is ready to receive funds. Once the target fund is collected, the SPV purchases the property and issues tokens proportional to each member's contribution. With their share of SPV tokens, the ivviator receives two specific rights of first refusal (ROFR) normally represented with two respective non-fungible tokens:

² The phrase "quarta via" is translated from Latin as "the fourth way."

- **ROFR to buy out:** The priority right to buy out ivviatees' tokens through periodic instalments (typically over a period similar to a mortgage, such as 20 years with buyouts on monthly basis). This creates an obligation for ivviatees to supply tokens for buyout, although each ivviatee retains the right to set the price for every deal. Additional rules are hardcoded into the smart contract to balance the deal's economy, further discussed in the section 'Balance of Interests'.
- **ROFR to lease:** The priority right to lease the property, intended to provide accommodation for the ivviator. If the ivviator exercises their leasing right, they will pay rent, minus their share in the property ownership, since the ivviator is essentially a co-owner and does not pay rent to themselves. If the ivviator decides not to take the lease, the property will still be rented out (as the SPV's investment property should generate income), and both the ivviator and the ivviatees will receive their respective shares of the rental income based on their token holdings in the SPV.



In summary, to ivviate a property means to purchase it through crowdfunding, leveraging financial technologies (Fintech) - tokens and smart contracts, and establishing a fund. The ivviator provides a down payment, while the ivviatees contribute the rest of the investment required to fund the property purchase. Once the property is transferred to the fund, the ivviator can gradually buy out the tokens at market price, exercising the right to become the sole owner. If the ivviator chooses not to buy out the tokens, they can be sold to others, who then become members of the fund. The ivviator may choose to live in the property, paying rent to the fund holding the house; otherwise, the property will be rented out to someone else. Once the property is fully paid off, the ivviator becomes its sole owner and can transfer the title to their name. As a result of the deal, real estate investors - ivviatees - enjoy monthly property appreciation, regularly selling a portion of their tokens. Unlike a mortgage, the ivviator can easily exit the arrangement by selling their share in the investment, transferring the rights to buy out and lease the property to a new beneficiary of the fund.

III. The Economy

A. Macroeconomy

Since the inception of the idea of tokenomics, there have been multiple concepts and attempts to

tokenise real estate. Several relatively successful implementations of this idea can be named. However, the article "Tokenisation of Real Estate: Evaluating the Promise of Securitisation" [1] asks why the idea has not expanded as some Fintech enthusiasts anticipated. The proposed research draws a parallel with real estate securitisation in traditional finance and highlights self-explanatory figures: only 1% of all real property is securitised and traded publicly, and this figure remains relatively stable. While the article does not find the answer as to why securitisation has not moved beyond a marginal phenomenon in the macroeconomy, it reasonably questions whether its alternative, i.e., tokenisation, can perform any better.

We do not claim that tokenisation is impossible. We are merely stating the fact, based on actual numbers, that the tokenisation process is not as widespread as some might have expected. It is hard to imagine a scenario where a random person walking down the street suddenly decides, out of the blue, to tokenise real estate. There must be clear economic incentives, not illusory ones, to tip the scales in favor of tokenisation, despite the fact that the industry is underdeveloped, lacks convenient services, is not cheap, not simple (for our random guy on the street), and lacks clear legal solutions.

The fourth approach appears to us as the driving mechanism of tokenisation. Tokenisation in this case is not the ultimate goal itself; it is genuinely a necessary tool to realise particular economic interests. This scheme could have been implemented with traditional financial instruments, and perhaps such examples can be found in the real economy. But with financial technologies, paper deals are replaced with the code of smart contracts on the blockchain, monthly meetings, and price negotiations - with dealings on a decentralised exchange that facilitates bargaining and settlement, immediately executing the exchange of tokens for digital payment - in cryptocurrency, stablecoins, and the like. All without the traditional intermediaries, e.g., financial brokers, escrow agents, lawyers, and so on. Fintech and Web3 provide the necessary tools that transform this scheme into a quick and accessible mechanism for implementing economic concepts without the costs of intermediaries.

Besides the two obvious groups of actors in this scheme—property buyers and investors—there is also a third group, i.e., exchanges and various DeFi platforms. They also become indirect beneficiaries of this system that will benefit from the inflow of liquidity, attracting more users to their ecosystems. Moreover, according to our observation, they urgently need additional liquidity and real applications of DeFi technology. The current DeFi is mainly composed of cryptocurrencies and stablecoins, where the former is highly volatile and hence can be extremely unstable, and the latter, being pegged to fiat currencies mostly USD or EUR, is highly liquid but not an investment in its nature but the medium of exchange. Tokenised real estate brings both components of moderate liquidity and profitability into the system.

B. Microeconomy

The mechanics of relationships in an inviated property are rather complex. Let us depict typical profiles of our participants. Alice is an individual who chooses to acquire a property without borrowing funds. It can also be a current mortgage payer seeking an opportunity to refinance or obtain relief from the debt. Her goal, similar to the majority entering the loan-mortgage system, is to own the property for residential purposes. There exists another group of borrowers—investors. While they also seek loans, their primary interest lies in the potential appreciation of property value, which they anticipate along with rent will offset the interest paid on the borrowed capital. Typically, both types of borrowers will make a down payment using their own funds. For instance, let us assume Alice has 20% of the

property's purchase price as her down payment. Similarly, Bob, Chuck, Dave, and Eve—who are property investors—each have 20% as well. Instead of borrowing and buying separate properties, in the ivvia scenario they can combine their resources to secure the full 100% required to purchase the property Alice desires.

The following calculations present two scenarios. The left part of the table illustrates a 20-year mortgage beginning in 2004 and ending in 2024. On the right, we explore the ivviate scheme, an alternative path Alice could have taken with the other investors. The full tables and methodology are added in the Annex.

Alice purchases a two-bedroom house on private land in Auburn³, a Middle-ring Sydney suburb, for \$520,000, the median price for such a house in the area at that time. Of this amount, \$416,000 is borrowed from a bank at an interest rate of 7.05%, according to the Australian Bureau of Statistics (ABS) for 'owner-occupier' loans. The interest rate is variable, and its changes are reflected in the table based on ABS data.

Initial house price: \$520K		
House price at closure (20 years): \$1.4M		
Average monthly payment: \$3K		
MORTGAGE	COMPARE	IVVIA
20 years	Payout term	15.5 years
\$866K	Total expenses	\$700K
\$533K	Investment Net Gain	\$1.2M

This forms the basis of our primary calculation assumption. Initially, Alice commits to a monthly mortgage payment of \$3,237.74, as per the annuity plan of her home loan. We will use this amount as a reference for what she allocates towards buying out her the ivviated property. This variable will be called “Ivvia Expenditure Cap” (IEC). The amount is adjusted each time the bank changes the interest rate in the mortgage scenario, ensuring the ivvia scheme matches the mortgage scenario's monthly commitments. For example, in the mortgage scenario, when the bank raised the interest rate to 7.30% in March 2005, Alice's monthly repayment increased to \$3,297.88. Consequently, Alice began allocating this adjusted amount towards buyouts in the ivvia scenario. Of course, the participants in

³ The suburb is randomly picked from the statistical data.

such a fund are free to agree on any alternative scheme. However, we use this parallel with the mortgage scenario to ensure comparability between the two.

Rent

However, \$3,237.74 Iovia Expenditure Cap is not the amount Alice can allocate towards monthly buyouts of her property share because it should be reduced by the cost of rent, as in mortgage she does not need to pay rent. Because for the ivviatees, it remains an investment property, they (the fund) will rent it to Alice. Statistics indicate that the median rent in that area was \$1,216.67 per month, which will be locked for each 12-month period⁴ under the ivvia. If rent increases significantly, as seen during certain periods (in 2009, 2010 and 2023), the rent is still capped at a 10% increase per year, in accordance with Australian law for ongoing leases. Initially, Alice owns 20% of the house, so she pays the ivviatees only \$973.33 (which the smart contract distributes according to their shares), and as her ownership share in the property grows, the rent she pays decreases accordingly.

Buyouts

Upon purchasing the property, the fund creates 300 tokens (with 8 decimals, i.e., 300.00000000), of which 60 are allocated to Alice as the owner of a 20% share. The remaining tokens are distributed among the other fund members based on their contributions: Bob, Chuck, Dave and Eve — 20% each.

Alice can allocate the difference between the Iovia Expenditure Cap (that is equal to mortgage repayment from the mortgage scenario) and her rent payment in a given month towards monthly buyouts. In the first month, the calculation is as follows:

$$\text{Iovia Expenditure Cap} - \text{Rent} = \text{Buyout Amount}$$

$$\$3,237.74 - \$973.33 = \$2,264.41$$

Since this follows the annuity scheme, similar to a mortgage, the number of tokens Alice purchases will be smaller initially but will steadily increase. The number of tokens she buys in a given month is determined by the current house price. While ivviatees are obligated to sell their tokens each month, each ivviatee is free to set the price for their share of tokens. Section "Balance of Interests" discusses extreme scenarios and how the proposed system balances them, and Section "Oracles" discusses the third party system that will feed market data on property prices and rent. At this stage, let us adhere to the official property value statistics in this area as our baseline scenario.

The next month after the purchase the statistic indicates the price growth, thus, \$520K turns into \$521K, which makes the price per token \$1736.67:

$$\frac{521,000.00}{300} = 1,736.67/\text{token}$$

⁴ Annual adjustments are based on ABS statistics, see Annex for methodological details.

For, the available \$2,264.41 Alice buys out 1.30388137 tokens from the ivviatees:

$$\frac{\$2,264.41}{\$1,736.67} = 1.30388137$$

These transactions occur monthly, based on three variables:

- House price with monthly adjustments.
- Rent price with annual adjustments.
- Interest rate, which is used as a reference to provide a comparable scenario to the mortgage.

All these adjustments enable Alice to gain full ownership of the property through ivvia in 185 months (15.5 years) compared to the basic 240-month (20 years) mortgage scenario, which is 38% faster than repaying the home loan.

Property total costs

From the demonstrated calculation we can summarise the following. If Alice and the rest of the fund had each pursued a mortgage separately, they would each have paid \$346K in total interest using the borrowed funds⁵, added to the initial house price \$520K the total expense gets to \$866K:

$$\begin{aligned} \text{House Price} + \text{Total Loan Interest} &= \text{Total Cost} \\ \$520,000.00 + \$346,125.23 &= \$866,125.23 \end{aligned}$$

In the ivvia scheme, we need to project net gains for the ivviator and ivviatees separately. Let us first take a look at Alice's benefits. She would pay off the property in 15.5 years, spending almost \$700K:

$$\begin{aligned} \text{Down payment} + \text{Total Buyouts} + \text{Total Rent} &= \text{Total Cost} \\ \$104,000.00 + \$501,993.31 + \$92,950.66 &= \$698,943.97 \end{aligned}$$

Comparing these two scenarios, we observe that Alice would pay significantly more in the home loan scenario: \$866K versus \$700K—a difference of \$166K, or 24% more for the same property.

⁵ For greater accuracy, however, we should increase the interest rate for investors since banks charge them more for mortgages than they do for owner-occupiers. But by doing so, the gap will widen even further, and the ivvia scheme for investors will become even more attractive.

	Mortgage	Ivvvia
Initial property price	\$520,000.00	
Property cost	\$866,125.23	\$698,943.97

Net gain

When comparing potential net gain—assuming that at the closure of the mortgage, Alice would sell the house—at first glance, the gain in the mortgage scenario could be \$345K greater based on ABS property pricing data, which values the house at \$1.4 million. Let us analyse the net gain assumptions for both scenarios. Under the mortgage, the net gain would be approximately \$533K over 20 years, and under the ivvvia - \$189K over 15.5 years:

$$\text{Property Value at Closure} - \text{Total Cost} = \text{Mortgage Net Gain}$$

$$\$1,400,000.00 - \$866,125.23 = \$533,874.77$$

$$\text{Property Value at Closure} - \text{Total Cost} = [\text{Alice's}] \text{Ivvvia Net Gain}$$

$$\$888,000.00 - \$698,943.97 = \$189,056.03$$

$$\text{Mortgage Net Gain} - [\text{Alice's}] \text{Ivvvia Net Gain} = \text{Difference}$$

$$\$533,874.77 - \$189,056.03 = \$344,818.74 \text{ (Nota bene! Incomparable)}$$

To make the ivvvia scenario comparable, we should assume Alice does not sell the property right after 15.5 years but instead holds it for 20 years to *match the mortgage scenario*. The net gain would be slightly above \$700K, which is \$167K more than the amount after the mortgage:

$$\text{Property Value at Closure} - \text{Down Payment} - \text{Total Buyouts} - \text{Total Rent} = \text{Ivvvia Net Gain}$$

$$\$1,400,000.00 - \$104,000.00 - \$92,950.66 - \$501,993.31 = \$701,056.03$$

$$\text{Ivvvia Net Gain} - \text{Mortgage Net Gain} = \text{Difference}$$

$$\$701,056.03 - \$533,874.77 = \$167,181.26 \text{ (Nota bene! Comparable)}$$

Comparison of Ivviatees' Benefits

Overall, the ivviatees' profit from their 80% contribution to the property purchase (\$416,000.00 out of \$520,000.00) amounts to \$178K over the entire period, and \$44K as their individual shares, since all equally co-invested 20%:

$$\text{Total Rent Revenue} + \text{Total Buyout Revenue} - \text{Ivviatees' Investment} = \text{Ivviatees' Profit}$$

$$\$92,950.66 + \$501,993.31 - \$416,000.00 = \$178,943.97$$

$$\text{Ivviatees' Profit} * \frac{\text{Individual Share}}{\text{Total investors' Share}} = \text{Individual Profit Share}$$

$$\$178,943.97 * \frac{20\%}{80\%} = \$44,735.99$$

However, to accurately compare individual ivviatees' benefits with those of mortgage investors, it is important to add one more variable: the potential investment accumulation of their funds over 20 years.

The profit in both scenarios is derived from different bases: in the mortgage scenario, it is the initial investment (down payment), followed by years of repayments which in fact are also investments, that are then subtracted from the property value at closure (assuming its sale). The net profit for ivviated property involves the initial investment (down payment) plus total rent revenue and total buyout revenue (according to each ivviatee's share). Thus, profit in the ivvia begins to generate from the first month through rent and sale margins, i.e., the difference between the initial purchase price and the buyout price. To make these two scenarios comparable, we need to consider the immobilisation of funds for monthly repayments in the mortgage scenario which are locked and would be paid off by the end of the mortgage term with property value appreciation. In contrast, the ivvia allows Bob to free up these funds, as there are no monthly commitments. Therefore, to compare these scenarios, we sum all repayments as if Bob had locked and accrued them in the ivvia scheme and add at least a bank deposit interest rate, assuming the money has been invested for 20 years, and then released similar to what happens after the house is sold when the mortgage is paid. The same applies to profits: if we wish to compare them to the mortgage where value is locked and appreciates after 20 years, our calculation should mimic this scenario. Thus, Bob would accrue profit from rent and buyouts and deposit them, along with the hypothetical "unspent mortgage repayments," into a bank deposit account (deposit accumulations are demonstrated in the second table in Annex):

$$\text{Mortgage Repayments} + \text{Rent Revenue} \times \text{Share} + \text{Buyouts Revenue} \times \text{Share} + \text{Interest} = \text{Net Gain}$$

$$\$762,125.23 + \$18,590.13 + \$100,398.66 + \$358,508.64 = \$1,239,622.66$$

Thus, Bob in the mortgage scenario would get \$533K while the ivvia offers \$1,24M through the span of 20 years, which is 140% bigger.

Resume

In summary, the ivviator benefits from buying out the tokenised property due to the reduced number of years required to fully own the house—15.5 instead of 20 years—and the lower cost Alice incurs in acquiring the property: \$700K compared to \$866K, which is \$166K cheaper. The ivviatee, under the same conditions as the mortgagor, benefits more from the ivvia, yielding \$750K more, i.e., \$1.24M vs. \$533K, while the ivviatee's profit, aside from the mortgage comparison will approximate \$44K, from the 20% share (\$104K) in the house investment.¹

Balance of Interests

This fund assumes that all parties act independently, each pursuing their own economic interests. The smart contract functionality provides economic liberty by allowing each ivviatee to sell their shares independently at their own chosen price. This approach aligns with the principle of decentralisation and free market - the core tenets of DeFi.

In the next section, we will discuss Oracles as a practical solution for providing data on market property prices and rent, which parties may choose to incorporate into their application. However, the primary idea is to preserve the necessary freedom for all participants to operate within the market. For example, Alice does not buy shares at a uniform price each month. Instead, she negotiates separately with Bob, Dave, and others. Each ivviatee is free to set their own price, while Alice is equally free to decide whether or not to buy.

Ivviatee's rights

The investment purpose of the fund for an ivviatee outlines several rules within the smart contract. First, Alice can purchase any number of tokens at any time, from the smallest unit to the entire holding of any given ivviatee. By setting his own price each month, Bob ensures that his investment remains profitable, whether it lasts 20 years or just one month. This arrangement also aligns with Alice's goal of acquiring the property. If she has the chance to make a full purchase earlier, rather than through monthly instalments, she may choose to buy it out completely.

If Bob wants to exit or partially cash out his investment, but Alice does not have the opportunity to buy the number of tokens Bob wishes to sell, he may offer them at auction following certain formalities. To initiate this, Bob must inform Alice of his intent to exit. Instead of the usual monthly buyout, where Alice has the right to purchase any number of tokens, Bob specifies the exact number of tokens he is offering and the price. Alice can either decline the offer, or it is considered declined if no response is given by the end of the month. After that, Bob is free to sell the tokens publicly, and the bulk of tokens will go to auction.

Hostile scenarios

Before we proceed to the auction as the core mechanism for reconciling opposing interests, let us highlight two crucial issues between the ivviator and ivviatee. In the previous section, we considered a favourable scenario where all parties acted rationally. But what happens if an ivviatee starts inflating the token price, making it impossible for Alice to ever buy out the property? A similar issue arises with

rent. What if the ivviatees set an unreasonably high rent? For example, if a hostile ivviatee sets the rent on their share so high that no one is willing to lease it, the entire fund suffers losses from unoccupied property.

Following the logic of DeFi, the system should not rely on any central authority to establish mandatory sale or rent prices. However, the absence of such oversight can potentially lead to conflicts and deadlocks unless the smart contract is properly designed to prevent this. There are two interrelated approaches to dealing with hostile actions related to buyouts and leases: buyouts become public, and rent is decided through voting, governed by specific rules.

Ivvia auction

The ivviatee's price offer is valid for one month, during which Bob can lower the price at any time if he wishes. However, by the end of the month, if Alice has not accepted the offer—meaning she has not purchased at least some of the tokens—their private relationship becomes public, i.e., the next month anyone can buy all the tokens in an auction. If the offered price is unacceptable to the ivviator, it will be determined by the open market. As mentioned earlier, the auction is also triggered if Alice declines to purchase a specific lot of tokens, in this case those tokens are auctioned. The goal is to provide liquidity by having the public market assess the property's value.

This type of auction is called an "English auction," a format familiar to many through platforms like eBay. Each participant submits their highest bid, which remains undisclosed. Instead, the price is incremented according to the auction rules—for example, by \$100 increments. The current price reflects the highest bid among all participants, but if someone places a much higher bid, the system will only increase the price by one increment (e.g., the current bid + \$100), holding the higher bid in reserve for further increments unless someone surpasses it.

For example, suppose there is a 10-token lot. Bidder 1 is willing to pay \$2,000 for the tokens, so the auction system shows the current price as \$100. Bidder 2 offers \$8,000, so the price increases to \$2,100, reserving Bidder 2's right to win unless someone outbids their maximum. Bidder 3 outbids Bidder 2 with a maximum bid of \$10,000. When the auction ends, Bidder 3 wins the lot for \$8,100.

There is also a reserve price mechanism for the ivviatee. If Bob wants to prevent his tokens from being sold at a price lower than he finds acceptable, he must bid himself and buy them back. This rule allows for a smooth process without imposing restrictive regulations, aligning with the free-market principle. The property's value will either be fairly determined at the auction, or, if Bob believes it is worth more than what buyers are willing to pay, he will need to buy it back at the price he deems fair. While this does not make hostile actions impossible, it requires significant capital to prevent Alice or anyone else from purchasing the tokens.

Ivviatee rent

The rent-balancing mechanism ensures that fund members cannot obstruct rent revenue streams and can get fair share of the rent. A house or apartment unit is considered an indivisible property. While there may be cases of leasing individual rooms or parts of a property, from the perspective of shareholders, this presents a problem. For instance, Bob's initial 20% share cannot be specifically allocated to a room or a kitchen—the property is indivisible. Therefore, unlike selling shares, leasing must be done in full, meaning all or nothing, which requires a collective decision-making process.

Two months before the current lease period (12 months) expires, any fund member (including the initiator) can propose a new rent price. During the following month, members vote based on their shareholdings (proportional voting). The proposal that receives 75% support is accepted. If no decision is made, the rent remains the same, and Alice has one month to either accept or decline it. The person who puts forward the proposal is not allowed to vote, ensuring that someone holding 75% of the shares cannot both offer the price and vote for it to manipulate the decision. This prevents Alice from depriving the invitees of rent revenue once her share reaches 75%, or an invitee to jack up the price.

Let us examine how this balances interests. Take a hypothetical scenario: as the buyout process nears its end, Bob has managed to retain his entire 20% stake by repeatedly buying back his tokens through the auction mechanism. Now, with only Alice and Bob left, Alice, despite holding more than 75% of the shares, cannot act unilaterally. Alice can propose a rent price, but Bob has no 75% of votes to agree to it. Therefore, only Bob can propose a rent price, which Alice can either accept or reject. If Bob does not propose a reasonable price, the rent remains unchanged until he decides to sell his property. They end up in a situation where Bob can keep the high price of his share for as long as he chooses to hold it, while Alice enjoys a fixed rent indefinitely, of which she only pays a fraction anyway. The remaining lease management decisions can be made by a simple majority vote, providing the necessary flexibility for casual issues.

Upon analysing this situation, we can conclude that there is no perfect solution. Imposing more elaborate rules and restrictions on Alice's or Bob's actions would undermine the free market concept and will also require a central authority deeply involved in arbitrating their economic strategies. Thus, a stalemate—where neither party can worsen the situation for the other—becomes the only viable way to balance the relationship. Alice will pay a fair rent and can sell her share on the open market. If Bob believes the auction does not offer a fair price for his share, he will wait until the market meets his expectations, though he will need to lock up his capital at the price he considers fair.

Centralisation

It is difficult to predict whether such a scheme can efficiently operate without a professional provider, because long-term economic relationships inevitably encounter a range of issues: bug fixes and smart contract upgrades, inheritance, and various unforeseen legal deadlocks or disputes. It may require third-party oracles to provide market data (as discussed further) and a professional manager, such as a real estate agent, to oversee the lease.

The smart contract will include an upgrade protocol⁶ that can be executed by fund members. It can be designed to accommodate different decision-making processes; however, the logic of this relationship suggests that any change should require unanimous agreement, as otherwise, it could leave room for collusion between the parties. However, in practice, unanimous agreement is difficult to achieve due to the often irrational behaviour of large social structures.

The scheme cannot scale effectively without proper mechanisms for professional management and arbitration, which inevitably require a central authority. Our intent as shown above is to leave as much economic freedom as possible, and design the smart contract in the way it accommodates a trusted third party authorised to intervene where there is no other way to sort out the issue. The ultimate

⁶ For example, using the ERC-1822 standard, known as the Universal Upgradeable Proxy Standard (UUPS).

resolution in deadlocks will depend on the conventional judicial system. If the parties are unable to resolve their disputes, suspect unfair actions, or face other issues, they can turn to the courts to obtain a legitimate decision. This decision would then be executed by the professional provider who holds the ultimate keys to their fund.

To ensure a smooth integration into the broader economy, the best approach is to operate IVVIA fund under a licensed financial provider, or—ideally, for market competition—through multiple independent providers. Therefore, our business model and commercial strategy, beyond demo projects and showcases, is built on a B2B approach, where we deliver robust technology to multiple market operators.

That said, we still believe in the possibility of a truly decentralised fund and aim to incentivise enthusiasts to ivviate property without intermediaries, as long as regulations permit. The core ivvia mechanism for private use will be available as open-source software, allowing anyone to use it at their own risk, without involving a trusted third party. Many jurisdictions, including Australia, provide regulatory exemptions for small-scale offerings, such as the "20/12/2" rule.

Closing notes on Implementation

Implementation involves two key elements: designing the application and adapting the ivvia economic model to fit specific legal jurisdictions. We have prepared a plan that addresses both aspects. The application will include a range of options to ensure compliance with various jurisdictions, while the second part of the plan focuses on launching the business in Australia, which serves as a model given the similarities in regulatory conditions across many countries.

Oracles

The part about balancing parties' interests presented the idealistic concept that aligns with principles of decentralisation and free market, we would like to emphasise that the presence of intermediaries is not a deviation from the principles. Third party services are called to make the user experience smoother. In a purely centralised system the absence of failure of the central party the relations tumble down. While decentralised systems withstand as they do not have one point of failure.

Oracles fill the gap of the missing element enabling the transition from the naive system presented above to the scalable application for mass market. Oracles are third party agents that will supply market data. Parties will be able to rely on current property prices and rent to develop their economic strategies and make decisions. Moreover, making it publicly, say Bob, connecting a price data provider to the smart contract makes his strategy available to anyone. His economic strategy that aligns with market trends becomes predictable for other DAO mates. Needless to say that such a system will require an "emergency brake" in case the oracle fails. To avoid a complete buyout for a fake price (supposeing oracle's system has been compromised) there will be a mechanism that will not let drain all the funds in one click, substantial transactions for more than 10% of the balance will require manual control of Alice and Bob.

Besides oracle, the apps will require functionality that make it possible to deal with a large number of participants. Say, Alice will need just to input a desired price cap to buyout all the matching offers of ivvatees, rather than being snowed under dealing manually with hundreds or thousands of offers every

month. Similar user-friendly functionality will need the process of offering and voting in the lease process.

Show case in Australia

In Australia, we plan to launch a showcase of the ivvia system by developing a Minimum ViableProduct (MVP). Partnering with a financial licence holder, or by obtaining our own Australian Financial Services Licence (AFSL), we will operate the scheme within a fully controlled, safe, and regulated environment.

As part of the contingency plan required by financial regulations, the fund will be capable of functioning entirely on paper if blockchain or the ivvia smart contract fails. This ensures that all transactions are backed by traditional legal frameworks and can independently function throughout the full lifecycle.

The advantage of this system lies in its ability to operate within the traditional financial environment while leveraging blockchain technology to enhance efficiency and scalability.

Annex:  IVVIA vs Mortgage